

#HW 6

● Graded

Student

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Total Points

100 / 100 pts

Question 1

1

50 / 50 pts

✓ - 0 pts correct

Question 2

2

50 / 50 pts

✓ - 0 pts Correct

- 2 pts $PV = mZRT/M$, so you should also include the molecular weight in the calculation. The remaining procedure is correct.

- 20 pts incomplete

- 25 pts late

No questions assigned to the following page.

$P_1 = 100 \text{ kPa} = 1 \text{ Bar}$, $T_1 = 300\text{K} = 26.85\text{C}$, $n = 1.25$

$P_2 = 2 \text{ MPa} = 20 \text{ Bar}$, $T_2 = ?$

Polytropic

$v_1 = (8.314 * 300 \text{ K} * 1000) / (100000 * 28.02) = 0.8901 \text{ m}^3/\text{kg}$

$Pv^{1.25} = Pv^{1.25}$ $v_2 = 0.08102 \text{ m}^3/\text{kg}$

$T_2 = (2000000 * 28.02 * 0.08102) / (8.314 * 1000) = 546.1\text{K}$

Specific Work (W/Mass) = $(P_2 * v_2 - P_1 * v_1) / (n - 1)$

$(2000000 * 0.08102 - 100000 * 0.8901) / (1.25 - 1) = 292120 \text{ J/kg}$

$T_2 = 546.1\text{K}$, specific work = 292.12kJ/kg

$P_c \text{ Nitrogen} = 33.9 \text{ bar or } 3.39 \text{ MPa}$ $P_{r1} = 0.0295$

$T_c \text{ Nitrogen} = 126\text{K}$ $T_{r1} = 2.38$

Z looks to be roughly 1.0, looking at $P_r < 1.0$

$P_{r2} = 0.590$ $T_{r2} = 4.33$ Z still looks to be 1.0

So the end result is going to end up the same, as Ideal $Z = 1$

AI seems to have trouble coming up with consistent answers, and does the problem differently as well, utilizing $PV=nRT$ to find T_2 as the Volumes cancel (somehow I still don't understand) and getting a final T of anywhere between 770-850 K the different times I asked chatGPT. After it uses specific work = $P_2V_2 - P_1V_1 / (1-n)$ which becomes $nR(T_2-T_1)/(1-n)$ and gets a final answer of -438kJ/kg

I don't really understand AI too well, but I do know it likes to "hallucinate" answers at times, so that could explain the discrepancy. my own answers didn't change as Z was very close to 1 anyways.

Ammonia production is heavily reliant on the Haber-Bosch process, which is a high PT process. An error in calculations could lower yields or cause accidents. It could also cost more money to design a system

No questions assigned to the following page.

$P_1 = 26 \text{ Bar}$, $T_1 = 298.15\text{K}$, $V = 50\text{m}^3$ or 50000 L

$P_{\text{Max}} = 201 \text{ Bar}$, $T_2 = 298.15\text{K}$, $T_d = 298.15\text{K}$, $T_{\text{out}} = 313.15\text{K}$

$$PV = nRT \quad 2600000 * 50 = n * 8.314 * 298.15$$

⑤ 25 Bar

$$n = 2600000 * 50 / (8.314 * 298.15) = 52444.3 \text{ moles of H}_2$$

$$50427.24 \text{ mole} * 2.0157 \text{ g/mole} = 105711 \text{ g H}_2$$

$Z =$ roughly 1.01 according to chart, $PV = nZRT$

$$2600000 * 50 = n * 1.01 * 8.314 * 298.15$$

$$n = 2.6\text{E}6 * 50 / (1.01 * 8.314 * 298.15) = 51925.1 \text{ moles H}_2$$

$$51925.1 * 2.0157 = 104665.4 \text{ g H}_2$$

a) 1.1E2 kg of H₂ b) 1.0E2 kg of H₂

⑥ 200 Bar

$$20100000 * 50 = n * 8.314 * 298.15$$

$$n = 20100000 * 50 / (8.314 * 298.15) = 405435.0 \text{ moles of H}_2$$

$$405435.0 \text{ mole} * 2.0157 \text{ g/mole} = 817227 \text{ g H}_2 - 105711 \text{ g H}_2$$

$Z =$ roughly 1.13, $PV = nZRT$

$$n = 2.01\text{E}7 * 50 / (1.13 * 8.314 * 298.15) = 358792.0 \text{ mole H}_2$$

$$358792.0 * 2.0157 = 723217.0 \text{ g H}_2 - 104665.4 \text{ g H}_2$$

a) 7.1E2 kg of H₂ b) 6.2E2 kg of H₂

$$1 - (6.2\text{E}2 / 7.1\text{E}2) = 0.139 \text{ or } 13.9\% \text{ error}$$

$$P * 50 = 405435.0 * 8.314 * 313.15$$

$$P = (8.314 * 313.15 * 405435.0) / 50 = 21111236 \text{ Pa} = 211 \text{ Bar}$$

$$Z = 1.13, P * 50 = 358792 * 1.13 * 8.314 * 313.15$$

$$P = (1.13 * 8.314 * 313.15 * 357792) / 50 = 21052394 = 211 \text{ Bar}$$

$Z =$ roughly 1.10 according to chart, $PV = nZRT$

$$20100000 * 50 = m * 1.10 * 8.314 * 313.15$$

$$n = 2.01\text{E}7 * 50 / (1.10 * 8.314 * 313.15) = 350922.3 \text{ mole H}_2$$

$$350922.3 * 2.0157 = 707354.0 \text{ g H}_2$$

a) 211 Bar, b) 211 Bar (Z should change here slightly, but not enough to alter the final answer within 2 sig figs), c) 7.1E2 kg

No questions assigned to the following page.

venting

At an average flow rate of 0.5kg/s, and running for 2 minutes.
 $T = 120s * 0.5kg/s = 60 \text{ kg of H}_2 \text{ released.}$

This much hydrogen released at once could result in a large combustion reaction that can burn/crush (explosive pressure) nearby people, and also suffocate them as well since H₂ combusting is $2(H_2) + O_2 = 2(H_2O)$ and consume oxygen too, not to mention H₂ is also an asphyxiant.

Prompt for 1a-b:

A fixed-roof hydrogen buffer tank upstream of a compressor is designed as follows:

- Internal volume: $V_{\text{tank}} = 50.0 \text{ m}^3$
- Maximum allowable working pressure (MAWP): 200 bar (gauge)
- PSV (pressure safety valve) set pressure: 200 bar (gauge)

· Ambient/design temperature: 25 °C
· Incoming gas temperature: up to 40 °C
Initially tank has dry Hydrogen at 25bar (g) and at 25°C.

Calculate the current mass of hydrogen in the tank:

(a) using the ideal gas law
(b) using the real-gas equation with an appropriate Z value.

Prompt for 2a-c:

Assuming ideal behavior and a final tank pressure of 200 bar (g) at 25 °C, calculate the maximum additional mass of hydrogen the operator believes the tank can receive.

Repeat step above using a compressibility factor Z appropriate for 200 bar. Compare the result to the ideal-gas calculation and find the percentage error.

Suppose the gas inside the tank later warms to 40 °C after filling.

(a) If the operator's calculated mass is added, what pressure will the tank reach (ideal-gas assumption)?
(b) What pressure will it reach with real-gas behavior?
(c) Solve for the safe mass of hydrogen that ensures the final pressure does not exceed 200 bar (g) at 40 °C using real-gas assumptions.

Prompt for Venting problem:

If the tank is overfilled and the PSV vents for 2 minutes at an average flow rate of 0.5 kg/s, estimate the total mass released. Briefly discuss why a hydrogen release of this magnitude is hazardous.

No questions assigned to the following page.

Write Up

The AI rounds in different spots, and also has trouble with keeping the digits correct, giving 1/10th the mass it would otherwise get for the first set. Given these are language models, and not calculators, I'd assume that is where the fault lies. This cascades further along, and the Z values are off too, again skewed by it being a language model and weird rounding.

Ideal gas VS Real gas was most noticeable at the 200 Bar problem, where Z was far off from 1.0

Ways to prevent disasters include relief valves, strict controls, fire suppressing, better designs (stronger), training, exhausts, safety redundancy, shutdown controls...

Understanding the importance in safety and design helps prevent serious disasters like chemical spills or explosions which can cause serious harm to the environment and people. Being able to see how to design limits and safety can help teach engineers how to safely create systems.